

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:BE01000111****Date:10-01-2025****Subject Name:Basic Electronics Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Explain working of Zener break down.	03
	(b) Explain forward bias PN junction diode with diagram.	04
	(c) Write a short note: V-I characteristic of P-N junction diode.	07
Q.2	(a) Why Zener diode can be used as voltage regulator?	03
	(b) Explain advantage and disadvantage of bridge rectifier over full wave rectifier.	04
	(c) Explain half wave rectifier with neat diagram and also draw the output waveform.	07
	OR	
	(c) Explain the shunt and series clipping circuits with diagram.	07
Q.3	(a) Draw the symbol of PNP transistor and NPN transistor and explain why it is called "Bipolar Transistor"?	03
	(b) Explain solar cell with diagram.	04
	(c) Draw the circuit of CB configuration of transistor and explain Input and output characteristics of CB configuration of transistor.	07
	OR	
Q.3	(a) Derive relationship between α dc and β dc of a transistor.	03
	(b) Explain fixed bias and self bias in a transistor.	04
	(c) Derive h parameters for transistor CE amplifier. Also Draw and explain h parameter equivalent model for CE amplifier.	07
Q.4	(a) Draw voltage multiplier circuit.	03
	(b) Explain FET as an Amplifier	04
	(c) Compare CE, CB and CC configuration with respect to different transistor.	07
	OR	
Q.4	(a) Explain in brief applications of JFET.	03
	(b) Explain Transconductance and switching in FET.	04
	(c) Compare the different characteristics of the following semiconductor devices: bipolar junction transistor, field-effect transistor.	07
Q.5	(a) Explain the AC load line of transistor.	03
	(b) Explain E-type MOSFET with neat diagram.	04
	(c) Write a short note on Tunnel diode.	07
	OR	
Q.5	(a) What is DC load line? Explain with necessary diagram.	03
	(b) Explain D-type MOSFET with neat diagram.	04
	(c) Write a short note on seven segment display with diagram.	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:BE01000111****Date:11-06-2025****Subject Name:Basic Electronics Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define the p-n junction diode and draw its characteristics.	03
	(b) Explain the working of a Zener diode as a voltage regulator.	04
	(c) Derive the expressions for load line analysis in a diode circuit.	07
Q.2	(a) Draw circuit diagram, waveform of a half-wave rectifier.	03
	(b) Explain one series and one shunt clipping circuits with necessary circuit diagram and waveform.	04
	(c) Explain the working of a full-wave bridge rectifier with necessary circuit diagram and waveform.	07
	OR	
	(c) Explain the working of a full-wave rectifier with a center-tapped transformer.	07
Q.3	(a) State the importance of transistor biasing.	03
	(b) Compare between CB, CE, and CC configurations in BJTs with diagram.	04
	(c) Draw and explain CE configuration with input and output Characteristics	07
	OR	
Q.3	(a) Derive relation between α and β .	03
	(b) Explain voltage divider biasing with neat diagram.	04
	(c) Draw and explain CB configuration with input and output Characteristics	07
Q.4	(a) Describe the operation of a JFET with neat diagram.	03
	(b) Draw MOSFET and describe the working.	04
	(c) Explain Voltage divider Biasing of JFET with diagram.	07
	OR	
Q.4	(a) State the advantages of JFETs over BJTs.	03
	(b) Draw symbol of MOSFETS. how enhancement MOSFET is different from depletion MOSFET.	04
	(c) Explain Self Biasing of JFET with diagram.	07
Q.5	(a) Illustrate LED.	03
	(b) What is a Schottky diode? Explain its advantages over regular diodes.	04
	(c) Discuss the construction and working of a solar cell. Highlight its applications.	07
	OR	
Q.5	(a) Explain Seven segment display.	03
	(b) Explain the working principle of a Tunnel diode.	04
	(c) Discuss the construction and applications of a Varactor diode	07

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000111/BE01R00111

Date: 30-12-2025

Subject Name: Basic Electronics Engineering

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) A BJT has emitter current and collector current of the value 10 mA and 9.8 mA respectively, determine current gains for CB and CE configurations.	03
(b) Explain forward and reverse bias operation of a P-N junction diode with depletion region.	04
(c) Draw CB configuration using NPN transistor and discuss its input and output characteristics with I_{CBO} , r_i , r_o , and current gain α .	07
Q.2 (a) Define knee voltage, bulk resistance and PIV for a P-N junction.	03
(b) Draw Bridge rectifier circuit and input and output waveforms. Also derive output voltages V_{dc} & V_{rms} .	04
(c) Draw circuit for voltage Doubler, Tripler & Quadrupler and explain their operation.	07
OR	
(c) Differentiate between Avalanche and Zener breakdown and discuss circuit of Zener as a voltage regulator.	07
Q.3 (a) Explain biased negative clipper with necessary waveforms.	03
(b) Discuss Choke input filter with its limitations.	04
(c) Discuss load line, operating point, DC & AC load lines for BJT.	07
OR	
(a) Describe positive clamper circuit with necessary waveforms.	03
(b) Describe RC filter and state ripple voltage formula.	04
(c) Discuss thermal stability of BJT with stability factors S , S' , S'' .	07
Q.4 (a) Explain FET as a switch.	03
(b) Draw Hybrid equivalent model of transistor for CB, CE, CC configurations.	04
(c) Explain CE amplifier with derivation of input impedance, output impedance and voltage gain in terms of h parameters.	07

OR

- (a) Explain FET as an amplifier. 03
- (b) Compare CB, CE, CC amplifiers. 04
- (c) Explain voltage divider bias circuit for BJT in detail. 07
- Q.5** (a) Discuss Solar cell in detail. 03
- (b) Explain Varactor diode with applications. 04
- (c) Explain Drain characteristics and Transfer characteristics of JFET in detail with all related terms as Transconductance g_m , drain resistance r_d , and amplification factor μ . 07

OR

- (a) Describe LED with applications. 03
- (b) Discuss Tunnel diode with its V/I characteristics and applications 04
- (c) Explain MOSFET and draw the structure for D-MOSFET & E-MOSFET. 07
